

M002 Week 4 — AND / OR

Topic: AND and OR Conditions · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

Last week you used **AND** (both true). This week you add **OR** (one is enough) and mix them.

- **AND** → runs **only when both** are true.
- **OR** → runs **when at least one** is true.

In Python: **and** , **or** . Detect a wall with `agent.detect(BLOCK, FORWARD)` .

1 · Predict 🧠

```
if wall ahead AND wall on right:
    turn left
```

Wall ahead, but the right is open. Does it turn left? Why?


```
if no wall ahead OR no wall below:
    move forward
```

Ahead is blocked, below is open. Does it move? Why?

2 · Trace It

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)  
elif not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

The agent has a wall in front and an open right. What runs?

The agent has open front, wall on right. What runs?

3 · Fill the Blanks

Turn left **only** at a corner (wall front AND wall right).

```
if agent.detect(BLOCK, FORWARD) ____ agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)
```

Move when **either** front OR below is open.

```
if not agent.detect(BLOCK, FORWARD) ____ not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

4 · Fix the Bug 🐛

The agent should move when **either** path ahead OR below is clear. This makes it wait for both.

```
if not agent.detect(BLOCK, FORWARD) and not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

Write the fixed line:

Why was it wrong?

5 · Finish + Show 📷

Go to your homework world. Solve the maze using **one AND** and **one OR**. Come back here.

Send a photo or video of the agent at the end. Write 3 or 4 lines.

I used **AND** when ...

I used **OR** when ...

One tricky part was ...

I fixed it by ...

Video plan. Film the maze on your phone. Plan your words first, then record.

1. Run your code from the start.
2. Read your **and** line and your **or** line out loud. Say what makes each run.
3. Say how **AND** is different from **OR**.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.